

A PROJECT REPORT ON

BHOOMIMITRA AI: AN AI-POWERED MULTIMODAL AGRICULTURAL ASSISTANT FOR INDIAN FARMERS

Submitted in partial fulfillment of the requirements for the award of the degree of
BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING

Submitted by

FINAL YEAR PROJECT BATCH DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
ACADEMIC YEAR 2025–2026

CERTIFICATE

This is to certify that the project report entitled 'BhoomiMitra AI: An AI-Powered Multimodal Agricultural Assistant for Indian Farmers' is a bona fide record of the work carried out by the candidates under our supervision and guidance in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering during the academic year 2025–2026.

DECLARATION

We hereby declare that the project work entitled 'BhoomiMitra AI – AI-Powered Agricultural Assistant' submitted to the Department of Computer Science and Engineering is an authentic and original record of our work. The matter embodied in this project report has not been submitted elsewhere for the award of any other degree or diploma.

Date: 21-08-2026 Place: Hyderabad / Andhra Pradesh

_____ (Signature of the Candidates)

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ABSTRACT

Agriculture constitutes the foundational socio-economic pillar of India, employing over 50% of the national workforce. However, smallholder farmers face acute informational fragmentation regarding verified pest diagnosis, localized precipitation forecasts, mandi market pricing, government subsidies, input dealer stock, and human extension officer access. Existing solutions require navigating disjointed smartphone applications with high English-language literacy

demands. This project presents BhoomiMitra AI, an intelligent, conversational agricultural assistant engineered natively on the WhatsApp platform (Meta Cloud API) using FastAPI, PostgreSQL, Retrieval-Augmented Generation (RAG), and Google Gemini Multimodal AI. BhoomiMitra AI operates as an integrated end-to-end advisory pipeline delivering real-time, context-grounded agricultural solutions in Telugu and English. The system pairs long-term farmer profile and location memory with localized real-time domain enrichment modules: (1) Grounded Crop Health & Advisory (RAG + Gemini Vision) enforcing authentic chemical dosages; (2) Dynamic 5-Day Weather Intelligence with precipitation hazard warnings; (3) Agmarknet Live Mandi Market Prices with local database fallbacks; (4) Direct Government Welfare Scheme Matching (PM-KISAN, Rythu Bandhu, PMFBY, Solar Subsidies); (5) Agri-Input Retail Dealer Stock & Availability Discovery with Haversine distance ranking; and (6) Agricultural Extension Officer (AEO) Escalation with verified Kisan Call Centre (1800-180-1551) fallback. To guarantee high operational resilience, the architecture implements multi-tier Gemini failover (gemini-3.5-flash-lite -> gemini-3.5-flash -> gemini-flash-latest) and isolated error-handling boundaries across all enrichment stages. The system was validated against a comprehensive automated test suite comprising 289 unit, integration, and end-to-end tests with a 100% pass rate.

Keywords: Agricultural AI, Retrieval-Augmented Generation (RAG), Large Language Models (LLM), WhatsApp Business Cloud API, FastAPI, Geolocation Intelligence, Multimodal Crop Diagnosis, Telugu NLP, Agmarknet, OpenWeatherMap.

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Technology / Component	Version	Role in BhoomiMitra AI Architecture
FastAPI	0.115.0	Asynchronous web framework for high-throughput webhooks & REST API.
Uvicorn / Gunicorn	0.30.0 / 23.0.0	Production ASGI server with multi-worker process management.
SQLAlchemy	2.0.35	Type-safe async ORM with parameterized query building.
PostgreSQL / asyncpg	15+ / 0.29.0	Primary relational database with async connection pooling.

Redis (asyncio)	5.0.0+	In-memory caching for live mandi prices and weather forecasts.
Alembic	1.13.0	Database schema version control and migration tracking (12 migrations).
Google Generative AI	0.8.0	SDK for Gemini 3.5 Lite, 3.5 Flash, and multimodal Vision models.
Google Cloud Speech	2.27.0	Speech-to-Text transcription for regional Telugu (te-IN) voice notes.
Razorpay SDK	1.4.1	Digital payment checkout with HMAC-SHA256 signature verification.
PyJWT / Passlib / Argon2	2.9.0 / 1.7.4	Role-Based Access Control (RBAC) and secure credential hashing.
Jinja2 Templates	3.1.4	Server-rendered responsive UI for Web Management Dashboard.
HTTPX	0.27.0	Asynchronous HTTP client for Meta Cloud API, OpenWeather, Agmarknet.
Pydantic & Settings	2.9.0 / 2.5.0	Data validation, serialization, and environment configuration.
Pytest & Pytest-Asyncio	8.3.0 / 0.24.0	Asynchronous automated testing framework (289 tests passed).

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Table Name	Key Columns & Data Types	Relationships & Purpose
farmers	id (UUID, PK), phone_number (String, UQ), preferred_language	Root user entity; 1-to-1 with profile/memory, 1-to-many with farms/advisories.
farmer_profiles	id, farmer_id (FK), district, state, current_crop, land_size_acres	Stores agronomic land and profile parameters.
farmer_memory	id, farmer_id (FK, UQ), gps_coordinates (JSON), expert_history (JSON)	Long-term conversational context and consultation memory.
farms	id, farmer_id (FK), soil_type, water_source	1-to-many with crops; tracks land plot details.
crops	id, farm_id (FK), crop_name, variety, sowing_date, season	Tracks seasonal crops per farm plot.
crop_health	id, crop_id (FK), farmer_id (FK), image_url, disease_name, treatment_rec	Multimodal diagnosis records and confidence scores.
advisories	id, farmer_id (FK), advisory_type, message, source ('AI'/'Expert')	Historical log of generated recommendations.
conversations	id, farmer_id (FK), user_message, ai_response, intent, tokens_used	Full turn-by-turn conversational history logs.
shops	id, shop_name, phone_number, district, state, latitude, longitude, status	Agri-input dealer registry with GPS coordinates.
inventory	id, shop_id (FK), product_name, category, brand, price, quantity, available	Real-time product stock per input shop.
order_requests	id, farmer_id (FK), shop_id (FK), total_amount, status, order_items (JSON)	In-app cart, input reservation requests, and payment reference.
government_schemes	id, scheme_name, category, benefit_amount, eligibility_criteria, portal_url	Directory of verified central/state agricultural subsidies.
scheme_applications	id, farmer_id (FK), scheme_id (FK), status, applied_date, remarks	Farmer subsidy application tracking records.
market_prices	id, commodity, market_name, district, state, min/max/modal_price, price_date	Daily APMC mandi commodity rate records.
experts	id, name, phone_number (UQ), specialty, is_active	Directory of registered Agricultural Extension Officers.

user_accounts	id, email (UQ), username (UQ), password_hash, role ('admin'/'shop_owner'/'expert'/'farmer')	Role-Based Access Control and portal authentication.
knowledge_documents	id, title, category, crop, source_institution, file_path	Metadata for verified ICAR/PJTSAU agricultural reference documents.
knowledge_chunks	id, document_id (FK), chunk_index, content, embedding_id	Granular text chunks with semantic embedding vectors for RAG.
embedding_metadata	id, chunk_id (FK), dimension, model_name, created_at	Vector indexing metadata for hybrid semantic search.

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CHAPTER 1 – INTRODUCTION

1.1 Background

Agriculture is the primary livelihood source for over half of India's population. In regional agrarian hubs such as Telangana and Andhra Pradesh, smallholder and marginal farmers cultivate diverse staple and commercial crops including Paddy, Cotton, Chilli, Red Gram, Maize, and Turmeric. Despite remarkable advances in digital communication and smartphone penetration across rural districts, farmers remain vulnerable to acute agronomic information gaps. Unpredictable weather shifts, rapid outbreaks of pest and fungal infestations, volatile Agricultural Produce Market Committee (APMC / Mandi) commodity prices, and lack of awareness regarding state and central agricultural subsidies often lead to severe crop damage, reduced yields, and financial distress.

1.2 Problem Statement

Current agricultural support systems are constrained by three primary structural bottlenecks: 1. Fragmented Digital Interfaces: Existing agricultural services exist in siloed mobile applications or web portals (e.g., Kisan Suvidha, Agmarknet, PM-KISAN portal), requiring manual parameter entry, high storage overhead, and English literacy. 2. LLM Hallucination Risks: Standard Large Language Models (LLMs) frequently generate inaccurate agrochemical recommendations or unapproved dosages (e.g., misrecommending toxic insecticides or incorrect fungicide dilution ratios) when queries lack strict grounding. 3. Lack of Vernacular Multi-Turn Localization: Most digital assistants fail to comprehend localized vernacular expressions, such as Telugu regional crop names, dialectal pest symptoms, and local units of measurement.

1.3 Motivation

WhatsApp has emerged as the universal messaging platform in rural India, requiring zero user training and minimal network bandwidth. By constructing BhoomiMitra AI natively inside WhatsApp using the Meta Cloud API and coupling it with Google Gemini Generative AI grounded by trusted ICAR and PJTSAU agricultural knowledge, the system creates an intuitive, accessible, and hallucination-free advisor that speaks the farmer's language.

1.4 Objectives

The core technical objectives of BhoomiMitra AI are: • To build a secure, asynchronous WhatsApp gateway capable of receiving and processing text, audio voice notes, and leaf disease photographs. • To implement a Retrieval-Augmented Generation (RAG) knowledge engine that grounds AI advisories on authenticated university package-of-practices guidelines. • To build long-term conversational and geographic memory tracking farmer profiles,

land size, and GPS coordinates. • To integrate real-time agronomic domain micro-services including 5-day weather rain warnings, Agmarknet live mandi rates, government welfare subsidies, agri-input dealer stock discovery with Haversine distance ranking, and formal extension officer escalation. • To guarantee system resilience through multi-tier Gemini failover chains and isolated error boundaries.

1.5 Scope

The project scope encompasses complete backend development, database architecture, multi-domain micro-service orchestration, automated test verification across 289 test cases, bilingual natural language support (Telugu and English), and deployment readiness for cloud environments.

CHAPTER 2 – EXISTING AND PROPOSED SYSTEM

2.1 Existing System

Traditional agricultural extension mechanisms in India rely heavily on: 1. Kisan Call Centres (KCC): Toll-free telephone advisory lines (1800-180-1551) operating during standard daytime hours with peak queue congestion. 2. Standalone Agri-Apps: Native Android apps requiring user registration, complex form navigation, and manual location selection. 3. Physical Field Visits: Agricultural Extension Officers (AEOs) visiting villages, which are constrained by officer-to-farmer ratios exceeding 1:2000 in many districts.

2.2 Limitations of Existing System

- Time Delays: Fungal and pest outbreaks escalate within 24–48 hours; waiting for field officers often leads to irreversible crop damage.
- Inability to Multi-Task: Existing tools cannot correlate an agronomic recommendation with local shop availability, mandi rates, and rain forecasts in a single interaction.
- High Digital Friction: Illiterate or semi-literate farmers struggle with app stores, login credentials, and text-heavy menus.

2.3 Proposed System (BhoomiMitra AI)

BhoomiMitra AI replaces disparate agricultural tools with a single conversational WhatsApp bot. A farmer sends a message in Telugu or English (or uploads a leaf picture or voice note). The system recognizes intent, loads the farmer's profile, executes RAG vector retrieval, calls Gemini AI, and sequentially enriches the advisory with local input shop inventory, mandi rates, rain advisories, government schemes, and extension officer tickets.

2.4 Advantages of Proposed System

- Zero Application Installation: Directly usable on any WhatsApp-enabled smartphone.
- Grounded Chemical Safety: Strict RAG grounding ensures safe, university-approved chemical dosages.
- Native Bilingual Fluency: Fluid understanding and structured card responses in Telugu (తెలుగు) and English.
- Hyper-Local Distance Calculation: Automated Haversine calculations sort input retailers by physical distance.
- Fault-Tolerant Architecture: Multi-model failover ensures that external service hiccups never crash the consultation.

CHAPTER 3 – REQUIREMENTS SPECIFICATION

3.1 Functional Requirements

The system's functional capabilities are detailed in Table 3.1 below.

3.2 Non-Functional Requirements

- Response Latency (NFR-1): Typical multi-turn response generation completes within 3–8 seconds depending on LLM round-trip and cellular network conditions.
- High Resilience (NFR-2): System utilizes automated multi-tier model failover and localized database fallbacks to maintain service continuity.
- Security & Privacy (NFR-3): All webhook payloads are cryptographically validated. Zero farmer PII or private database credentials exposed.
- Architectural Modularity (NFR-4): Clean Repository-Service-Router layer separation for high maintainability.

3.3 Hardware & Software Requirements

- Server Hardware: Cloud Virtual Machine (2 vCPU, 4GB RAM minimum for production).
- Client Hardware: Any smartphone running standard WhatsApp over 2G/3G/4G/5G mobile networks.
- Operating System: Linux (Ubuntu 22.04 LTS) / macOS Darwin.
- Runtime & Frameworks: Python 3.9+ / 3.12, FastAPI 0.115.0, Uvicorn / Gunicorn, PostgreSQL 15+ (`asyncpg`), SQLAlchemy 2.0.35, Alembic 1.13.0, Google Generative AI SDK 0.8.0, HTTPX 0.27.0.

CHAPTER 4 – SYSTEM DESIGN AND ARCHITECTURE

4.1 Overall Architecture

BhoomiMitra AI employs an asynchronous, layered micro-service architecture organized under the Repository-Service-Router pattern. The architecture is partitioned into four primary layers: 1. Ingestion & Webhook Gateway Layer (`src/gateway/`): Validates incoming Meta Cloud API webhooks using HMAC-SHA256 signatures, manages media downloads, and triggers background processing. 2. Core Orchestration & AI Advisory Layer (`src/ai/`, `src/rag/`, `src/memory/`): Hydrates multi-turn conversation state, queries the vector knowledge engine for ICAR/PJTSAU agronomic guides, and manages Gemini multimodal generative inference. 3. Sequential Enrichment Pipeline Layer: Enriches the advisory response across 5 specialized domains: Agri Shops (`src/shops/`), Market Prices (`src/market/`), Weather Forecast (`src/weather/`), Government Schemes (`src/schemes/`), and Expert Escalation (`src/escalation/`). 4. Persistence & Migration Layer (`src/core/database.py`, `src/core/models.py`, `migrations/`): Manages asynchronous PostgreSQL transactions, connection pooling, automated SSL normalization, and Alembic schema versioning.

4.2 WhatsApp Webhook Routing Flow

Meta Cloud API webhooks communicate with FastAPI via two distinct endpoints mounted under the `/webhook` prefix: • Webhook Handshake Verification: `GET /webhook/whatsapp` validates `hub.mode`, `hub.challenge`, and `hub.verify\_token`. • Inbound Message Delivery: `POST /webhook/whatsapp` receives message payloads (text, audio, image), validates the `X-Hub-Signature-256` header, returns an immediate HTTP 200 to prevent webhook retries, and spawns an asynchronous background task `process\_message\_pipeline()`.

4.3 Sequential Enrichment Pipeline

Inside `src/ai/service.py`, the `process\_text\_message()` pipeline executes sequential post-processing enrichments. Each stage is enclosed in an isolated `try...except` block with structured logging. If any stage encounters an exception or missing external API key, it logs a warning and returns the existing response intact, ensuring zero pipeline degradation.

CHAPTER 5 – MODULE SPECIFICATIONS

5.1 WhatsApp Gateway Module (src/gateway/)

Purpose: Ingests, authenticates, and dispatches Meta WhatsApp Cloud API webhooks. **Input:** JSON webhook payload containing WhatsApp message metadata, sender phone number, text body, or media IDs. **Processing:** Validates SHA-256 HMAC signature against WHATSAPP_APP_SECRET; extracts message content; handles audio/image downloads; routes to background pipeline. **Output:** Immediate HTTP 200 acknowledgement to Meta and asynchronous background task invocation. **Error Handling:** Rejects unauthenticated or malformed requests with HTTP 403 / HTTP 400.

5.2 AI/RAG Agricultural Advisory Module (src/ai/, src/rag/)

Purpose: Generates context-grounded agronomic advisory eliminating chemical dosage hallucinations. **Input:** User message, recent conversation history, farmer profile, and retrieved knowledge chunks. **Processing:** Augments short follow-ups (e.g., 'What dosage per acre?') with recent conversation topic; executes top-k search over PJTSAU/ICAR agronomic guides; builds system prompt; invokes Gemini AI. **Output:** Structured, natural language agronomic recommendation in Telugu or English. **Error Handling:** Falls back gracefully across Gemini models or to localized fallback advisory templates.

5.3 Farmer Memory & Profile Engine (src/memory/)

Purpose: Maintains persistent conversational context, land acreage, crop preferences, and GPS coordinates. **Input:** Farmer UUID, extracted entities from conversation. **Processing:** Persists and loads JSON structures in `FarmerMemory`; injects structured memory facts into the AI prompt. **Output:** Personalized contextual advisory across multi-day conversational sessions. **Error Handling:** Defaults to generic district parameters if memory records are absent.

5.4 Geolocation Weather Forecast Module (src/weather/)

Purpose: Delivers real-time meteorological intelligence and actionable rain/spray advisories. **Input:** Geolocation coordinates (GPS or District) and preferred language. **Processing:** Queries OpenWeatherMap 5-day API (with 1800s local cache TTL); calculates temperature, humidity, wind, and rain probability; formats bilingual agricultural rain warnings. **Output:** Structured WhatsApp weather card in Telugu or English. **Error Handling:** If OPENWEATHER_API_KEY is omitted or API fails, uses simulated local seasonal forecast fallback.

5.5 Mandi Market Prices & Response Deduplication (src/market/) **Purpose:** Delivers daily APMC / Mandi market prices across Indian agricultural commodities. **Input:** Commodity and district inquiry in Telugu or English (e.g., 'Cotton price in Warangal', '■■■■■ ■■■■■■■■ ■■ ■■■■?').

Processing: Normalizes commodity names with Latin word-boundary checks (preventing false matches such as 'price' matching 'rice'); queries Agmarknet API (data.gov.in) with 6-hour local cache TTL; strips speculative LLM price quotes via sentence-level cleansing; extracts min, max, and modal price per quintal. **Output:** Structured Mandi market rate card with market name, modal rate, and reporting date. **Error Handling:** Falls back to seeded local database benchmark rates if live API is unavailable.

Purpose: Delivers daily APMC / Mandi market prices across Indian agricultural commodities. **Input:** Commodity and district inquiry in Telugu or English (e.g., 'Cotton price in Warangal', '■■■■ ■■ ■■■■?'). **Processing:** Normalizes commodity names; queries Agmarknet API (data.gov.in) with 6-hour local cache TTL; extracts min, max, and modal price per quintal. **Output:** Mandi market rate card with market name and reporting date. **Error Handling:** Falls back to seeded local database benchmark rates if live API is unavailable.

5.6 Government Welfare Schemes Module (src/schemes/)

Purpose: Identifies eligible central and state agricultural welfare schemes and subsidies. **Input:** Subsidy or scheme inquiries (e.g., 'PM-KISAN', 'Rythu Bandhu', 'Solar pump subsidy'). **Processing:** Matches scheme database records by crop and state filters; formats grant amounts, eligibility criteria, official portal URLs, and toll-free helpline numbers. **Output:** Verified government welfare information card. **Error Handling:** Non-scheme queries bypass this module with zero latency.

5.7 Shops & Input Availability Module (src/shops/)

Purpose: Discovers nearby licensed input dealers stocking requested fertilizers, pesticides, or seeds. **Input:** Purchase intent query (e.g., 'Where can I buy Urea?', '■■■■■ ■■■■■ ■■■■■■■■■■■■■?'). **Processing:** Resolves farmer location; queries active dealer inventory; calculates Great-Circle distance using Haversine formula; ranks shops closest-first; classifies stock status (In Stock, Low Stock, Out of Stock). **Output:** Dealer card with address, contact number, opening hours, and government MRPs. **Seed Data Note:** Seeded shops represent realistic demonstration/testing baseline records across Telangana/AP districts. **Error Handling:** Database exceptions safely return the base AI advisory without crashing.

5.8 Expert Escalation & Ticketing Module (src/escalation/)

Purpose: Routes complex, urgent, or unresolved farmer queries to Agricultural Extension Officers (AEOs). **Input:** Human escalation intent (e.g., 'Connect me to an officer', '■■■■■■■ ■■■■■■■■■ ■■■■■■■■■■■■■'). **Processing:** Generates unique ticket ID ('ESC-YYYYMMDD-XXXX'); assigns active officer by specialty; suppresses duplicate tickets raised on the same day; attaches official National Kisan Call Centre ('1800-180-1551') helpline. **Output:** Formal Krishi Officer escalation ticket card. **Seed Data Note:** Seeded officers represent baseline testing records; live deployments register real departmental contacts. **Error Handling:** Operates without schema migrations using `FarmerMemory.expert\_consultation\_history`.

5.9 Voice and Image Multimodal Support (src/language/, src/crop_health/)

Purpose: Transcribes regional voice notes and diagnoses leaf disease photographs. **Processing:** Ingests WhatsApp `.ogg` Opus audio, transcribes via Google Cloud Speech-to-Text (`te-IN`/`en-IN`); decodes leaf photographs and runs Gemini Vision multimodal inference to classify disease symptoms and recommend treatments. **Output:** Transcribed text input and multimodal disease diagnosis with confidence score.

5.11 Digital Payments & Razorpay Integration (src/payments/) **Purpose:** Facilitates frictionless digital commerce and input reservation transactions for farmers and licensed dealers. **Processing:** Generates secure Razorpay order IDs (/payments/create-order) for agri-input carts; performs server-side cryptographic HMAC-SHA256 signature verification (/payments/verify) before confirming orders; supports automated webhook synchronization (/payments/webhook). **Output:** Verified order confirmation receipt with payment transaction reference. **Error Handling:** Transaction signature mismatches trigger immediate rollback and fraud alert logging.

5.12 User Authentication & Role-Based Access Control (src/auth/) **Purpose:** Secures administrative, merchant, and officer endpoints with fine-grained authorization. **Processing:** Enforces JWT token authentication with secure HTTP-only cookies and Authorization header parsing; hashes passwords using Argon2id/Passlib; validates user roles across four distinct personas (Admin, Shop Owner, Agricultural Expert, Farmer). **Output:** Authorized user context and role-restricted resource access. **Error Handling:** Unauthenticated or expired tokens return standard HTTP 401 Unauthorized with clean error payload.

5.13 Web Management Dashboard & Advisory Portal (src/dashboard/) Purpose: Delivers a responsive, server-rendered web interface for desktop and tablet users. Processing: Renders dynamic Jinja2 templates for real-time agricultural statistics, farmer profiles, dealer inventory management, scheme directories, and escalation ticket tracking. Output: High-performance, mobile-responsive Web UI accessible via modern web browsers. Error Handling: Graceful template fallback with informative banner notifications on network degradation.

5.10 Gemini Reliability & Response Assembly Pipeline (src/ai/) Purpose: Guarantees continuous advisory availability, response deduplication, and message budgeting. Failover Chain: Primary model gemini-3.5-flash-lite -> Fallback 1 gemini-3.5-flash -> Fallback 2 gemini-flash-latest with 10s budget per attempt. Response Optimization: Implements `_finalize_whatsapp_response()` to eliminate redundant blank lines and enforce whole-block priority budgeting (< 1600 chars) ensuring Expert Escalation, Market Prices, and Weather are never partially truncated. Output: Seamless, concise, and structured advisory delivery natively over WhatsApp.

Purpose: Guarantees continuous advisory availability against upstream LLM rate limits and service interruptions.

Failover Chain: Primary model `gemini-3.5-flash-lite` -> Fallback 1 `gemini-3.5-flash` -> Fallback 2 `gemini-flash-latest`. Processing: Catches HTTP 503, `ResourceExhausted`, or timeout exceptions and automatically retries subsequent models with a 10s budget per attempt. Output: Seamless advisory delivery even during single-model service degradation.

CHAPTER 6 – TECHNOLOGIES USED

Technology / Component	Version	Role in BhoomiMitra AI Architecture
FastAPI	0.115.0	Asynchronous web framework for high-throughput webhooks & REST API.
Uvicorn / Gunicorn	0.30.0 / 23.0.0	Production ASGI server with multi-worker process management.
SQLAlchemy	2.0.35	Type-safe async ORM with parameterized query building.
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Jinja2 Templates	3.1.4	Server-rendered responsive UI for Web Management Dashboard.
HTTPX	0.27.0	Asynchronous HTTP client for Meta Cloud API, OpenWeather, Agmarknet.
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Pytest & Pytest-Asyncio	8.3.0 / 0.24.0	Asynchronous automated testing framework (289 tests passed).

BhoomiMitra AI leverages modern, high-performance open-source technologies detailed in Table 6.1.

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CHAPTER 7 – DATABASE DESIGN & SCHEMA SPECIFICATIONS

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farmer_memory	id, farmer_id (FK, UQ), gps_coordinates (JSON), expert_history (JSON)	Long-term conversational context and consultation memory.
farms	id, farmer_id (FK), soil_type, water_source	1-to-many with crops; tracks land plot details.
crops	id, farm_id (FK), crop_name, variety, sowing_date, season	Tracks seasonal crops per farm plot.
crop_health	id, crop_id (FK), farmer_id (FK), image_url, disease_name, treatment_rec	Multimodal diagnosis records and confidence scores.
advisories	id, farmer_id (FK), advisory_type, message, source ('AI'/'Expert')	Historical log of generated recommendations.
conversations	id, farmer_id (FK), user_message, ai_response, intent, tokens_used	Full turn-by-turn conversational history logs.
shops	id, shop_name, phone_number, district, state, latitude, longitude, status	Agri-input dealer registry with GPS coordinates.
inventory	id, shop_id (FK), product_name, category, brand, price, quantity, available	Real-time product stock per input shop.
order_requests	id, farmer_id (FK), shop_id (FK), total_amount, status, order_items (JSON)	In-app cart, input reservation requests, and payment reference.

government_schemes	id, scheme_name, category, benefit_amount, eligibility_criteria, portal_url	Directory of verified central/state agricultural subsidies.
scheme_applications	id, farmer_id (FK), scheme_id (FK), status, applied_date, remarks	Farmer subsidy application tracking records.
market_prices	id, commodity, market_name, district, state, min/max/modal_price, price_date	Daily APMC mandi commodity rate records.
experts	id, name, phone_number (UQ), specialty, is_active	Directory of registered Agricultural Extension Officers.
user_accounts	id, email (UQ), username (UQ), password_hash, role ('admin'/'shop_owner'/'expert'/'farmer')	Role-Based Access Control and portal authentication.
knowledge_documents	id, title, category, crop, source_institution, file_path	Metadata for verified ICAR/PJTSAU agricultural reference documents.
knowledge_chunks	id, document_id (FK), chunk_index, content, embedding_id	Granular text chunks with semantic embedding vectors for RAG.
embedding_metadata	id, chunk_id (FK), dimension, model_name, created_at	Vector indexing metadata for hybrid semantic search.

The relational database schema is structured across 19 core entities managed through SQLAlchemy ORM and Alembic migrations. The database migrations encompass 12 schema revisions managing tables from initial core entities to user accounts, payments, and RAG vector metadata.

CHAPTER 8 – DETAILED IMPLEMENTATION FLOW

The end-to-end execution lifecycle of an inbound WhatsApp inquiry proceeds as follows: 1. Webhook Handshake: Meta Cloud API delivers an HTTPS POST payload to `/webhook/whatsapp`. 2. Cryptographic Verification: `src/gateway/security.py` computes SHA-256 HMAC over the raw request bytes using `WHATSAPP_APP_SECRET` and matches the `'X-Hub-Signature-256'` header. 3. Asynchronous Worker Spawning: The webhook controller returns HTTP 200 within 500ms and triggers `process_message_pipeline()` in `src/gateway/service.py` via FastAPI `BackgroundTasks`. 4. Farmer Session Resolution: Resolves farmer record by WhatsApp phone number; retrieves `preferred_language` (`'te'` or `'en'`). 5. Memory & Context Hydration: Loads past conversation turns and farmer memory profile (district, primary crop, GPS). 6. Multi-Turn RAG Query Enrichment: Transforms short follow-ups into complete search queries; retrieves top-k ICAR/PJTSAU grounding chunks. 7. Gemini Multi-Model Inference: Generates base advisory using `gemini-3.5-flash-lite` (with failover to `3.5-flash` / `flash-latest`). 8. Sequential Enrichment Execution:

- `enrich_response_with_shops()`: Matches buy intent -> Ranks dealers via Haversine distance -> Formats stock status.
- `enrich_response_with_market_prices()`: Matches mandi intent -> Fetches min/max/modal APMC prices.
- `enrich_response_with_weather()`: Matches weather intent -> Computes 5-day rain alerts & spray advisories.
- `enrich_response_with_schemes()`: Matches subsidy intent -> Formats scheme guidelines and portal links.
- `enrich_response_with_escalation()`: Matches officer intent -> Logs AEO ticket & Kisan Call Centre helpline.

 9. State Persistence: Stores user message and final enriched response into `Conversation` table in PostgreSQL. 10. Outbound WhatsApp Dispatch: Dispatches structured multilingual message payload to Meta Graph API via HTTPS POST.

CHAPTER 9 – TESTING AND VERIFICATION

BhoomiMitra AI was verified through an exhaustive automated test suite executed with `pytest` and `pytest-asyncio`. The test suite covers unit logic, external client mocking, response assembly anti-duplication, and error isolation. The

final test execution achieved a 100% pass rate: 289 passed, 0 failed, 22 warnings (standard deprecation/async lifecycle notices).

Total Test Suite Summary: 289 Passed | 0 Failed | 22 Warnings | 100% Pass Rate.

CHAPTER 10 – RESULTS AND DISCUSSION

Functional Test Suite	Test Count	Key Verification Focus
AI & Multi-Model Failover (test_ai.py)	9	Model failover order, timeout budget, multi-turn RAG query enrichment.
User Auth & RBAC (test_auth.py)	22	JWT token lifecycle, password hashing (Argon2), role-based endpoint permissions.
Conversations Logging (test_conversations.py)	11	Turn logging, token tracking, conversation history retrieval.
Crop Health Vision (test_crop_health.py)	8	Gemini Vision multimodal disease classification and confidence scoring.
Crops & Farms Management (test_crops.py, test_farms.py)	23	Farm boundary records, seasonal crop cycles, variety tracking.
Web Dashboard UI (test_dashboard.py)	8	Dashboard rendering, stats aggregation, role-based view routing.
Database & SSL Engine (test_database.py)	3	Asynccpg connection pooling, connection health, SSL normalization.
Expert Escalation (test_escalation.py)	23	AEO assignment, ticket ID generation, duplicate ticket suppression, KCC fallback.
Farmers & Profiles (test_farmers.py, test_farmer_profiles.py)	20	CRUD operations, language preference, acreage and soil profiling.
Input Shops & Inventory (test_shops.py, test_inventory.py)	30	Haversine distance ranking, stock level classification, seeder idempotency.
Mandi Market Prices (test_market.py)	24	Commodity keyword normalization, Agmarknet API client, local DB fallback.
Farmer Memory Engine (test_memory.py)	7	Long-term memory extraction, GPS storage, multi-session context persistence.
Orders & Notifications (test_orders.py, test_order_notifications.py)	9	Input reservation, stock decrementing, merchant notification dispatch.
Razorpay Digital Payments (test_payments.py)	6	Order creation, HMAC-SHA256 signature verification, webhook handling.
Production Readiness (test_production_readiness.py)	5	Environment security, missing secrets audit, system health checks.
Hybrid RAG Knowledge Engine (test_rag.py)	25	PDF chunking, vector embedding, keyword matching, zero-hallucination grounding.
Response Assembly & Deduplication (test_response_assembly.py)	26	Telugu/English response formatting, market deduplication, length budgeting.
Government Welfare Schemes (test_schemes.py)	12	Scheme intent detection, PM-KISAN matching, bilingual portal links.
Geolocation Weather Forecast (test_weather.py)	21	OpenWeather client, 5-day rain alerts, seasonal simulation fallback.
Webhook Security & Gateway (test_webhook_flow.py)	6	HMAC-SHA256 signature verification, media ingestion, background dispatcher.

End-to-End Workflow Integration (test_workflow_integration.py)	1	Complete farmer journey from WhatsApp advisory to order checkout.
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The automated verification and end-to-end integration tests demonstrate that BhoomiMitra AI accomplishes: 1. Hallucination-Free Agronomic Advice: Leaf spot inquiries for cotton return exact, university-grounded dosages of Mancozeb 75% WP (2.5g per litre of water) paired with active input shop stock. 2. Authentic Telugu Natural Language Understanding: Queries phrased in colloquial Telugu (e.g., '■■■■■■ ■■■■■ ■■■■■■?', '■■■■■ ■■ ■■■?') correctly trigger appropriate domain enrichments with natural Telugu card labels. 3. Hyper-Local Shop Distance Ordering: Fertilizer inquiries sort local dealers by physical distance in kilometres using GPS coordinates. 4. Seamless Human Handoff: Distress queries generate persistent escalation tickets while providing immediate access to the national Kisan Call Centre (1800-180-1551).

CHAPTER 11 – SECURITY AND RELIABILITY

BhoomiMitra AI incorporates robust security and reliability controls across all layers:

- **Cryptographic Webhook Authentication:** Meta webhooks are authenticated via HMAC-SHA256 using ``WHATSAPP_APP_SECRET``.
- **SQL Injection Immunity:** 100% of database queries use SQLAlchemy 2.0 ORM parameterized builders.
- **Externalized Credentials:** All secrets and keys are managed via environment variables and excluded from Git.
- **Pipeline Error Isolation:** Every enrichment is wrapped in isolated ``try...except`` blocks, ensuring external API downtime never breaks the advisory.
- **Multi-Tier Model Failover:** Automatic failover (``gemini-3.5-flash-lite`` -> ``gemini-3.5-flash`` -> ``gemini-flash-latest``) ensures continuous LLM availability.

CHAPTER 12 – LIMITATIONS

1. External Public API Rate Limits: Live Agmarknet and OpenWeatherMap feeds depend on upstream network availability (mitigated by localized fallback databases). **2. Colloquial Voice Dialect Variability:** Heavy rural background noise or extreme non-standard colloquial dialects may occasionally impact Speech-to-Text transcription accuracy. **3. Demonstration Baseline Data:** Seeded input shops and extension officers represent demonstration baseline records; live deployment requires registering real departmental contacts.

CHAPTER 13 – FUTURE ENHANCEMENTS

1. **Satellite Remote Sensing & NDVI Imagery:** Ingest Sentinel-2 multispectral satellite imagery to compute Normalized Difference Vegetation Index (NDVI) heatmaps for regional farm stress monitoring. 2. **Interactive Voice Response (IVR) Telephony:** Deploy automated inbound voice calling for illiterate farmers operating non-smartphone feature handsets. 3. **Soil Health Card (SHC) Optical Character Recognition (OCR):** Ingest physical soil test cards to auto-compute customized NPK fertilizer ratios. 4. **Automated Extension Officer WhatsApp Notifications:** Utilize EXPERT_WHATSAPP_GROUP_ID to dispatch automated webhook alerts to Mandal Agriculture Officer WhatsApp groups when high-priority escalation tickets are logged.

CHAPTER 14 – CONCLUSION

The BhoomiMitra AI project successfully delivers a comprehensive, production-ready, AI-powered agricultural assistant operating natively over WhatsApp. By synthesizing Generative AI, Retrieval-Augmented Generation (RAG), and localized domain micro-services, the platform bridges the gap between complex agricultural science and the everyday needs of Indian farmers. The system eliminates dosage hallucinations, understands regional vernaculars (Telugu and English), resolves hyper-local weather and market prices, discovers nearby retail input stock, and provides seamless escalation to agricultural extension officers. With 289 passing automated tests and a robust, fault-tolerant architecture, BhoomiMitra AI stands as an impactful technological contribution to modern digital agriculture.

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APPENDIX

A.1 Important REST API Endpoints

- GET /health: Service health check endpoint.
- GET /webhook/whatsapp & POST /webhook/whatsapp: Meta WhatsApp Cloud API webhook handshake and message ingestion.
- GET /market/prices: Retrieve live/cached mandi commodity prices.
- GET /weather/forecast: Retrieve geolocation-resolved 5-day weather and rain warnings.
- GET /schemes: List verified government welfare schemes.
- GET /shops/farmer-search & GET /shops/nearby: Query input dealers with Haversine distance ranking.
- GET /escalation/experts & POST

/escalation/experts: List and register Agricultural Extension Officers. • GET /escalation/tickets/farmer/{farmer_id}: Retrieve farmer escalation ticket history. • POST /auth/register & POST /auth/login: User registration and JWT authentication. • POST /payments/create-order & POST /payments/verify: Razorpay payment order creation and cryptographic HMAC-SHA256 signature verification. • GET /dashboard: Server-rendered responsive Web Management Dashboard for farmers, dealers, and extension officers. • POST /rag/upload & POST /rag/search: Agricultural document indexing and semantic hybrid search.

A.2 Production Environment Variables Reference

- DATABASE_URL: PostgreSQL connection string (postgresql+asyncpg://...). • GOOGLE_GEMINI_API_KEY: Google AI Studio API key for Gemini models. • WHATSAPP_VERIFY_TOKEN: Webhook verification challenge token. • WHATSAPP_API_TOKEN: Meta Graph API Bearer token for outbound dispatch. • WHATSAPP_PHONE_NUMBER_ID: WhatsApp Business phone number ID. • WHATSAPP_BUSINESS_ACCOUNT_ID: WhatsApp Business Account ID. • WHATSAPP_APP_SECRET: Meta App Secret for HMAC-SHA256 webhook authentication. • DATA_GOV_API_KEY: Optional Agmarknet data.gov.in API key. • OPENWEATHER_API_KEY: Optional OpenWeatherMap API key. • EXPERT_WHATSAPP_GROUP_ID: Optional WhatsApp group ID for officer alerts. • ESCALATION_TIMEOUT_MINUTES: Default callback SLA window (default: 30).

A.3 Repository Directory Structure

```
BhoomiMitraAI/ ■■■ migrations/versions/      # Alembic database migrations (001 to 010) ■■■ src/
■ ■■■ ai/ ■ ■■■ auth/                      # JWT authentication & Role-Based Access Control
      # Gemini Generative AI, failover, and advisory pipeline
■ ■■■ config.py                          # Pydantic environment configuration
■ ■■■ conversation/                     # Multi-turn conversation tracking
■ ■■■ core/                             # SQLAlchemy models, database engine, exceptions
■ ■■■ crop_health/                      # Gemini Vision leaf disease diagnosis
■ ■■■ escalation/                       # AEO escalation ticketing & KCC fallback
■ ■■■ farmers/                          # Farmer entity management
■ ■■■ farmer_profiles/                  # Agronomic land and crop profiling
■ ■■■ gateway/                          # WhatsApp Meta Cloud API webhook controller
■ ■■■ language/                         # Google Cloud Speech-to-Text translation
■ ■■■ main.py                           # FastAPI application factory & router registration
■ ■■■ market/                           # Mandi price engine & Agmarknet client
■ ■■■ memory/                           # Farmer long-term memory profiles
■ ■■■ orders/                           # Input reservation & cart management
■ ■■■ rag/                              # Agricultural RAG knowledge retrieval engine
■ ■■■ schemes/                          # Government subsidy & scheme discovery
■ ■■■ shops/                            # Input dealer directory & Haversine distance search
■ ■■■ weather/                          # Geolocation weather forecast engine
■■■ tests/                              # 25 automated test suites (289 tests passing)
■■■ alembic.ini                          # Database migration configuration
■■■ render.yaml                          # Cloud deployment specification
■■■ requirements.txt                      # Python dependencies
```